

# Chapter 5, Section 3

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1. Let  $X$  be a nonsingular projective variety of any dimension, let  $Y$  be a nonsingular subvariety, and let  $\pi: \tilde{X} \rightarrow X$  be obtained by blowing up  $Y$ . Show that  $p_a(\tilde{X}) = p_a(X)$ .

*Proof.* We can generalize the proof of (3.4).

**Proposition 0.1.** *In the above setting, we have  $\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ , and  $R^i \pi_* \mathcal{O}_{\tilde{X}} = 0$  for  $i > 0$ . Therefore,  $H^i(X, \mathcal{O}_X) \cong H^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$  for all  $i \geq 0$ .*

By (II, 8.24), the strict transform of  $Y$  in  $\tilde{X}$  is isomorphic to the projection space bundle  $\mathbb{P}(\mathcal{I}/\mathcal{I}^2)$ , where  $\mathcal{I}$  is the ideal sheaf  $\mathfrak{f}$ . It is locally free of rank  $r = \text{codim}(Y, X)$  since  $Y$  is nonsingular. In particular,  $\mathbb{P}(\mathcal{I}/\mathcal{I}^2)$  has dimension  $\dim Y + \text{codim } r - 1$ , so it corresponds to a divisor on  $X$ . By (II, 7.15), there exists a commutative diagram

$$\begin{array}{ccc} \tilde{Y} & \xrightarrow{\tilde{f}} & \tilde{X} \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X \end{array}$$

where  $f$  is a closed immersion, and so is  $\tilde{f}$ . By the theorem on formal functionns, For any scheme theoretic point  $y \in Y$ , the natural map

$$\hat{\mathcal{F}}^i_y = (R^i \pi_* \mathcal{O}_{\tilde{X}})_y \otimes \hat{\mathcal{O}}_{y, Y} \cong \varprojlim H^i(Y_n, \mathcal{O}_{Y_n}).$$

By (7.11),  $\pi_* \mathcal{O}_{\mathbb{P}(\mathcal{I}/\mathcal{I}^2)} = \mathcal{O}_X$

Since  $\pi$  is an isomorphism of  $V = \tilde{X} - E$  onto  $U = X - Y$  (II, 7.13). The sheaves  $\mathcal{F}^i = R^i \pi_* \mathcal{O}_{\tilde{X}}$  for  $i > 0$  have support at  $Y$ .  $\square$

2. Let  $C$  and  $D$  be curves on a surface  $X$ , meeting at a point  $P$ . Let  $\pi: \tilde{X} \rightarrow X$  be the monoidal transformation with center  $P$ . Show that  $\tilde{C} \cdot \tilde{D} = C \cdot D - \mu_P(C) \cdot \mu_P(D)$ . Conclude that  $C \cdot D = \sum \mu_P(C) \cdot \mu_P(D)$ , where the sum is taken over all intersection points of  $C$  and  $D$ , including infinitely near intersection points.
3. Let  $\pi: \tilde{X} \rightarrow X$  be a monoidal transformation, and let  $D$  be a very ample divisor on  $X$ . Show that  $2\pi^*D - E$  is ample on  $\tilde{X}$ .
5. Let  $a_1, \dots, a_r$ ,  $r \geq 5$ , be distinct elements of  $k$ , and let  $C$  be the curve in  $\mathbb{P}^2$  given by the (affine) equation  $y^2 = \prod_{i=1}^r (x - a_i)$ . Show that the point  $P$  at infinity on the  $y$ -axis is a singular point. Compute  $\delta_P$  and  $g(\tilde{Y})$ , where  $\tilde{Y}$  is the normalization of  $Y$ . Show in this way that one obtains hyperelliptic curves of ever genus  $g \geq 2$ .
7. For each of the following singularities at  $(0,0)$  in the plane, give an embedded resolution, compute  $\delta_P$ , and decide which ones are equivalent.
  - (a)  $x^3 + y^5 = 0$
  - (b)  $x^3 + x^4 + y^5 = 0$
  - (c)  $x^3 + y^4 + y^5 = 0$
  - (d)  $x^3 + y^5 + y^6 = 0$
  - (e)  $x^3 + xy^3 + y^5 = 0$